Name ____ Date ___________ Period _____

Transformations of f(x) = |x|, Part I

Algebra II Honors | Unit 2 | Day 2 | TEK A2.6.C

Objective: I can describe, graph, and justify the effect of a and b on f(x) = a|bx|, including why some transformations produce no visible change.

Vocabulary

Vertical stretch/compression — af(x); |a|>1 stretches away from the x-axis, 0<|a|<1 compresses toward it.

Horizontal stretch/compression — f(bx); |b|>1 compresses toward the y-axis, 0<|b|<1 stretches away from it.

Invariant point — a point that does NOT move under a transformation. For f(x) = a|bx|, the ___________ is always invariant.

Part 1 — The Effect of a and b, Combined

Function	a	b	Effect
f(x) = 2|x|	2	1	________________________
f(x) = (1/2)|x|	1/2	1	________________________
f(x) = |2x|	1	2	__________________________
f(x) = -2|3x|	-2	3	____ ___

Part 2 — Why the Vertex Never Moves

Show algebraically that (0, 0) is on the graph of f(x) = a|bx| no matter what a and b are:

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f(0) = a|b · 0| = a|0| = a · ______ = ______ → the vertex (0, 0) is _______________ under every a and b

Part 3 — Why Reflecting Over the y-axis Does Nothing

f(x) = |-x| should reflect the parent over the y-axis (b = -1). Prove algebraically that it produces the SAME graph as f(x) = |x|:

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|-x| = |-1| · |x| = __________________________ → identical to the parent - this is the same even-function fact from Day 1 (f(-x) = f(x))

Part 4 — Which Graphs Are Really the Same?

For each pair, determine whether the two functions produce the SAME graph. Justify using the properties of absolute value.

1. f(x) = 3|x| and f(x) = |3x| ___________ since |3x| = |3||x| = 3|x|
2. f(x) = -2|x| and f(x) = 2|-x| _________________ -2|x| opens downward; 2|-x| = 2|x| opens upward
3. f(x) = (1/2)|4x| and f(x) = 2|x| ___________ since (1/2)|4x| = (1/2)(4)|x| = 2|x|
4. f(x) = |6x| and f(x) = 6|x| ___________ since |6x| = |6||x| = 6|x|

Part 5 — Going Further: Factoring Out a Constant

Tomorrow's lesson adds shifts: f(x) = a|x - c| + d. Some functions LOOK like they have a b-value inside, but a constant can be factored out to reveal the real a. Try it:

f(x) = 2|3x - 6| = 2|3(x - 2)| = 2 · ______ · |x - 2| = ______|x - 2| → the REAL a-value is ______

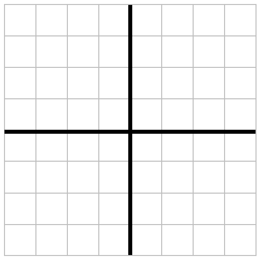
You Try. Factor $f(x) = 4|2x + 8|$ to find the real a-value and the vertex's x-coordinate.

$f(x) = 4|2(x + 4)| =$ ______ $|x + 4|$ → $a =$ ______ vertex x-coordinate ________

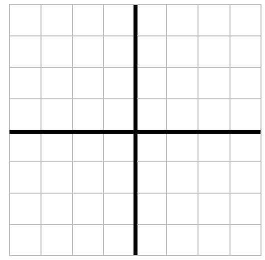
Part 6 — Graph and Compare

Graph both on their own grid. Each square is 1 unit; both axes run -5 to 5.

A. $f(x) = 2|x|$



B. $f(x) = (1/2)|x|$



Part 7 — Practice

1. Factor $f(x) = 3|6x + 18|$ into $a|x - c|$ form and give the real a-value. ______________________

2. Order from flattest to steepest: $|x|$, $3|x|$, $(1/4)|x|$, $-2|x|$. ______________________________

3. True or False: $f(x) = |-5x|$ and $f(x) = |5x|$ produce different graphs. Justify.

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4. Two functions, $f(x) = a|x|$ and $g(x) = |bx|$, look identical for every x. What relationship must a and b satisfy?

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5. Factor $f(x) = 7|-2x + 8|$ into $a|x - c|$ form and give the real a-value and vertex x-coordinate.

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Exit Ticket

Are $f(x) = -3|x|$ and $f(x) = 3|-x|$ the same graph? Justify algebraically, then describe every transformation present. Finally, factor $f(x) = 5|4x - 20|$ to find its real a-value.

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